Master's Thesis

Evaluating the Posterior Calibration and Trustworthiness of Generative Molecular Docking Models

M.Phil. Data Intensive Science

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Abstract

Molecular docking predicts how a small molecule binds to a target protein, a key early step in drug discovery. Recent generative deep-learning models have advanced the field rapidly, yet they are judged almost entirely by how often their single best-ranked pose is correct, and whether the generated poses are physically valid, overlooking whether the model's distribution of poses reflects honest uncertainty rather than overconfidence.

This study asks whether such models can be trusted beyond accuracy, and what determines that trustworthiness. We adapt two likelihood-free calibration tests, TARP and MIRA, to docking and evaluate two state-of-the-art models, DiffDock and SigmaDock, across datasets, protein families, and the inference-time sampler, including deterministic versus stochastic dynamics and its number of steps.

We find that physical validity is set by the generative architecture, not the post-hoc ranking stage; that both models are systematically overconfident, independently of accuracy and validity. The calibration is governed primarily by the architecture, with the sampler a genuine secondary lever. A ten-step stochastic sampler gives SigmaDock its best calibration and fastest runtime at minor accuracy cost. Trustworthiness in generative docking is thus invisible to accuracy-only evaluation, motivating calibration-aware model and sampler selection.

Contents

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1 Introduction

Molecular docking is a fundamental computational technique in structure-based drug discovery and the study of biological processes. Its goal is to predict how a small molecule (ligand) binds a target macromolecule (receptor or protein) from their unbound structures. By predicting both the bound conformation (pose) and the binding affinity, docking lets researchers screen vast virtual libraries for therapeutic leads [1].

As Stärk et al. [2] summarise, drug discovery takes about a decade, costs on the order of a billion dollars, and can fail at any stage. Additionally, there exist an estimated 10^{60} drug-like molecules [3], which implies a vast search space and is far beyond the reach of experimental structure determination [4].

Recent progress has pivoted toward deep generative models, which reframe docking as sampling from a learned conditional distribution $p(\theta \mid x)$ [2, 5, 6, 7] rather than optimising a physics-based scoring function [1, 8]. These methods report striking gains in benchmark-averaged accuracy, which says little about whether a model can be trusted on a specific receptor. Their posteriors are, in fact, frequently overconfident [9]. In such a high-stakes, failure-prone pipeline, a confidently wrong pose is far more damaging than an honestly uncertain one, which makes trustworthiness, and not only headline accuracy, the decisive question.

Accuracy-based evaluation has a structural blind spot that surfaces in two ways. First, the metric itself: reporting only the displacement (measured via root-mean-squared-deviation; RMSD) between predicted and crystal poses can obscure true predictive ability, since deep-learning methods competitive on RMSD frequently produce physically invalid poses [10]. Second, and more fundamentally, both RMSD and physical validity are properties of a single point prediction. Neither asks whether the *distribution* of sampled poses carries the right uncertainty, whether the model assigns correct probability mass to its learned conditional distribution [11, 12]. This distributional calibration is the property we add to the evaluation, enabling us to measure the reliability of docking models.

Assessing calibration this way requires likelihood-free diagnostics, since generative docking models admit no tractable likelihood. Earlier simulation-based tools [13, 9] depend on a chosen credible region or scaling poorly to high dimensions, making them inadequate for docking. The Test of Accuracy with Random Points (TARP) [11] and the recent MIRA statistic [12] circumvent this, and enable the measuring of conditional-distribution accuracy and Bayesian model comparison, while transferring reliably to generative models [14].

A second lever on the same blind spot concerns not what a model learns but how it is sampled: the reverse-time integrator that turns a trained score into a pose is chosen entirely at inference time and, as in image diffusion [15, 16, 17], shapes sample quality as much as the architecture does. This has scarcely reached docking, where DiffDock and SigmaDock both default to first-order Euler schemes and the sampler's effect on accuracy, and the learned posterior is uncharacterised. Crucially, this is invisible to accuracy alone: as we show, top-1 accuracy is nearly unchanged across samplers and step counts, even where calibration is not.

These blind spots motivate a departure from accuracy-centric evaluation. This thesis asks whether the learned posteriors of generative docking models are trustworthy beyond headline accuracy, and whether that trustworthiness is determined by the model's architecture or by the sampler used to integrate it. By trustworthiness we mean the reliability of a model's posterior beyond point accuracy: whether its sampled poses are physically valid (PoseBusters) and whether their distribution is calibrated (TARP and MIRA). To answer this, we make two contributions. First, we adapt the likelihood-free TARP and MIRA diagnostics to the docking manifold, quantifying posterior calibration alongside physical validity and accuracy. This investigation asks whether validity is fixed by a model's geometric inductive biases or recoverable by a post-hoc ranking. Second, we open the sampling design space: the reverse stochastic differential equation (SDE) against the probability-flow ordinary differential equation (ODE), and the number of discretisation steps. This study on sampling, subject to posterior fidelity that accuracy-only tuning leaves unexamined, highlights the need for the introduced metric. Together, these let us attribute reliability to its architectural and inference-time sources.

The remainder of the thesis is organised as follows. **Section 2** reviews related work, and **Section 3** the diffusion, TARP, and MIRA framework with the DiffDock and SigmaDock formulations. **Section 4** covers the benchmarks, metrics, and test adaptations. The results are presented in **Section 5**, including the validity, calibration, and sampler results, and **Section 6** concludes.

2 Related Work

We situate our work at the intersection of structure-based drug discovery, which supplies the task and its benchmarks, and generative modelling, which supplies the methods now redefining it. We first organise docking tasks and the classical and learned model families that address them (Sect. 2.1–2.2), then review the two lines of research that frame our contributions: the likelihood-free evaluation of generative *posteriors* (Sect. 2.3) and the diffusion *sampling* design space (Sect. 2.4).

2.1 Docking Tasks and Paradigms

Docking tasks are categorised, both mathematically and practically, by the level of prior knowledge, the algorithm is given about the target's structure, which sets the size of the search space and must be kept in mind when comparing benchmarks:

- **Redocking and virtual screening:** In industrial lead optimisation and standard virtual screening, the binding site is usually known in advance, often from a holo-crystal structure. The task is constrained to predicting the ligand's orientation and affinity within a predefined pocket, prioritising throughput and exact chemical validity [7]. Search-based methods and SigmaDock are designed for this area [7].
- **Blind and apo docking:** Apo docking has grown in importance alongside protein-structure predictors such as AlphaFold [18], where the bound structure is unknown. The algorithm must evaluate the entire receptor surface to locate orthosteric or allosteric sites, a substantially harder problem and one where DiffDock excels [6].
- **Joint docking:** Most models treat the protein as rigid, which is generally false for side chains. The most recent paradigm folds the protein simultaneously around a conformationally flexible ligand [19, 20].

The algorithms addressing these tasks fall into three classes:

- **Search-based methods:** The models decouple docking into a stochastic search over poses scored by a physics- or energy-based function, including AutoDock Vina [1], Gold [21], Glide [8], and, more recently, the runtime-efficient Gnina [22].
- **Regression-based methods:** Those methods bypass iterative search with a one-shot prediction of the complex, using a trigonometry-aware network [5] or an $SE(3)$-equivariant graph neural network [2] trained under an MSE objective.
- **Generative methods:** Recent models reframe docking as sampling from a learned conditional distribution, which avoids the average-pose collapse that an MSE loss induces on a multimodal target [6].

A comprehensive review of the deep-learning transformation of docking, its shortcomings, and emerging solutions, comparing 27 models, is given by Lee et al. [23].

2.2 Generative Docking and Physical Validity

DiffDock [6] established the generative paradigm by framing docking as a diffusion process over ligand degrees of freedom, sidestepping both the steric clashes of regression models and the runtimes of search-based ones. Its successor DiffDock-L [24] improved generalisation through model scaling, synthetic data, and a confidence-bootstrapping training loop, and is the model we evaluate. A growing family has followed, including the physics-augmented DiffDock-Glide [25] and the fragment-based SigmaDock [7].

A persistent criticism, however, is physical validity. Buttenschoen et al. [10] showed with the PoseBusters suite that deep-learning methods, which look competitive on RMSD frequently generates physically invalid poses, and that performance degrades sharply on targets dissimilar to the training set. Two responses have emerged: post-hoc correction, such as energy minimisation or molecular-dynamics alignment [26, 25], and architectural inductive biases that constrain generation to physically permissible structures, the route taken by SigmaDock [7]. The mathematical formulations of DiffDock and SigmaDock, on which our calibration analysis rests, are developed in Sect. 3.3 and Sect. 3.4.

2.3 Evaluating Generative Posteriors

A parallel question is how the distribution produced by such samplers should be judged. Reporting top-1 RMSD (Sect. 4.2) measures a single point and is silent on whether the model's uncertainty is faithful. Work in simulation-based inference has shown that learned posteriors are frequently overconfident [9], motivating dedicated calibration diagnostics: simulation-based calibration [13], expected-coverage tests, and the Test of Accuracy with Random Points (TARP) [11], have been shown to transfer to generative models [14]. The recent MIRA statistic [12] adds a scalar measure suited to Bayesian model comparison. Adapting these likelihood-free tests to the docking manifold and using them to compare generative docking models, is the first contribution of this thesis. Their derivations are set out in Sect. 3.5.

2.4 Sampling and the Diffusion Design Space

Generative docking models inherit not only the architecture of diffusion models but also their inference machinery, which in the broader literature has proven at least as consequential: a single trained score can be integrated in many ways, decoupling sampling from training [27, 15]. Deterministic few-step and higher-order solvers [28, 16, 29, 30] and the deterministic–stochastic trade-off [17] together form a design space that drives much of the gain in sample quality and efficiency observed in image-generation, which is further developed in Sect. 3.2.2. Despite this, the design space has scarcely been examined in docking: both DiffDock [6] and SigmaDock [7] default to first-order Euler integration, and the effect of the sampler on the learned *posterior*, as well as top-1 accuracy, remains uncharacterised. Quantifying that effect is the second contribution of this thesis.

3 Theoretical Background

Denoising Diffusion Probabilistic Models (DDPMs) are latent-variable models that learn to synthesise data by inverting a fixed corruption process [31, 27]. This Section develops the diffusion framework underpinning generative docking, following Luo [32], and then specialises it to the DiffDock and SigmaDock formulations and the TARP and MIRA calibration tests used in our evaluation.

3.1 Molecular Docking

Statistically, molecular docking is the task of modelling the highly multimodal conditional distribution $p(\theta \mid x(P, L))$, where the random variable θ collects the ligand degrees of freedom (translation, rotation, and torsion angles) given the crystallised complex x, consisting of the protein P and the ligand chemical graph L [6]. As motivated in Sect. 2, framing this distribution generatively avoids the average-pose collapse of regression models and the runtimes of search-based docking. The remaining part of this Section develops the diffusion machinery that makes it computationally feasible.

3.2 Forward and Reverse Diffusion Process

A diffusion model defines a forward Markov chain that progressively degrades D-dimensional data $\mathbf{x}_0 \in \mathbb{R}^D$ into Gaussian noise over T steps under a fixed variance schedule $\beta_1, \ldots, \beta_T$ [31]. Its defining property, introduced by Ho et al. [27], is that with $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$ the marginal is available in closed form, so any timestep can be sampled directly and training stays tractable:

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}\big(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0,\, (1-\bar{\alpha}_t)\mathbf{I}\big), \qquad \mathbf{x}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \tag{3.1}$$

Generation inverts this chain from a standard-normal prior $p(\mathbf{x}_T) = \mathcal{N}(\mathbf{0}, \mathbf{I})$. Rather than predicting the reverse-transition mean directly, Ho et al. [27] train a network $\boldsymbol{\epsilon}_\theta$ to predict the injected noise, which reduces the evidence lower bound to a simple denoising objective:

$$\mathcal{L}_{\text{simple}}(\theta) = \mathbb{E}_{t,\mathbf{x}_0,\boldsymbol{\epsilon}}\left[\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}, t)\|^2\right]. \tag{3.2}$$

3.2.1 Connection to Score-matching and continuous-time SDE

The noise-prediction parameterisation [27] bridges DDPMs and score-based generative modelling [33]. Regressing a denoiser onto the added noise is, up to a scale factor, denoising score matching [34], which avoids the second-derivative term of the original objective [35]. The score can therefore be read directly from the learned noise prediction,

$$\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t) \approx -\frac{\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)}{\sqrt{1-\bar{\alpha}_t}}, \tag{3.3}$$

This defines a vector field pointing toward regions of high likelihood and equivalently by Tweedie's formula [36], the score is the scaled residual between the denoised estimate and the current state (Eq. 3.7). Song et al. [15] unified these views in continuous time: as $T \to \infty$ the forward chain converges to an Itô SDE,

$$\mathrm{d}\mathbf{x} = \mathbf{f}(\mathbf{x}, t)\,\mathrm{d}t + g(t)\,\mathrm{d}\mathbf{w}, \tag{3.4}$$

with drift $\mathbf{f}(\mathbf{x}, t)$, diffusion coefficient $g(t)$ setting the variance schedule, and Brownian motion $\mathbf{w}$. Sampling simulates the reverse-time SDE [37], which depends only on the marginal score $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$,

$$\mathrm{d}\mathbf{x} = [\mathbf{f}(\mathbf{x}, t) - g(t)^2\,\nabla_{\mathbf{x}} \log p_t(\mathbf{x})]\,\mathrm{d}t + g(t)\,\mathrm{d}\mathbf{w}, \tag{3.5}$$

and Song et al. [15] showed that the deterministic probability-flow ODE shares the same marginals as this reverse SDE:

$$\mathrm{d}\mathbf{x} = \left[\mathbf{f}(\mathbf{x}, t) - \tfrac{1}{2}g(t)^2\,\nabla_{\mathbf{x}} \log p_t(\mathbf{x})\right]\mathrm{d}t. \tag{3.6}$$

Both the reverse SDE and the ODE are used in the inference and coverage analysis. Because the learned score is a fixed map of the perturbed data space, it does not depend on how we sample from it: score estimation is decoupled from the forward trajectory, and continuous-time models can be integrated with diverse solvers at inference time.

3.2.2 Sampling the Learned Score

To navigate these geometry-aware trajectories, models rely on iterative numerical solvers rather than single-step predictions: a one-shot prediction from pure noise averages across all candidate pockets and yields a blurry, uncommitted pose, whereas iterative refinement lets the network commit progressively to a specific, high-density binding mode [6, 16].

Both DiffDock and SigmaDock default to first-order Euler schemes. For the probability-flow ODE, the Euler method follows the score over an interval Δt and, assuming a constant score across the step, accumulates a truncation error of $\mathcal{O}(\Delta t)$. For the SDE, the Euler–Maruyama method [38] adds to this drift a scaled Wiener term $g(t)\sqrt{\Delta t}\,z$, injecting Gaussian noise at each step to maintain exploration. From a noise step, the model outputs the learned score

$$s_\theta(x_t, t) = \frac{\hat{x}_0 - x_t}{\sigma_t^2}, \tag{3.7}$$

with $\hat{x}_0$ the predicted denoised sample (Tweedie's formula, [36]), giving the two update rules:

$$x_{t-1} = x_t + g^2 \cdot s_\theta \cdot \Delta t + g \cdot \sqrt{\Delta t} \cdot z \qquad \text{(SDE)} \tag{3.8}$$

$$x_{t-1} = x_t + \tfrac{1}{2} \cdot g^2 \cdot s_\theta \cdot \Delta t \qquad \text{(ODE)} \tag{3.9}$$

To mitigate this linear error scaling without inflating the step count, the diffusion design space adopts deterministic few-step and higher-order schemes, such as the Heun predictor–corrector [16] and DPM-Solver [29, 30], which reach high-order accuracy in as few as 10–20 steps. Beyond efficiency, the solver also shapes the sampled distribution and hence calibration. This is governed by a trade-off between two error sources [17]. Deterministic ODE integration minimises discretisation error but, lacking a corrective term, behaves as a mode-seeking map, whereas the stochastic SDE injects a Langevin correction that contracts the error of an imperfect score [15] at the cost of more steps. Neither dominates, which motivates the SDE/ODE comparison central to this work.

3.2.3 Diffusion Models on Riemannian Manifolds

The continuous SDE framework is built for unbounded Euclidean space, yet a docking pose lives on a constrained geometry: describing it via independent Cartesian coordinates is inefficient and risks violating rigid constraints such as fixed bond lengths and stereochemistry. De Bortoli et al. [39] established that the diffusion generative framework transfers to Riemannian manifolds, where the true and parameterised scores are elements of the tangent space, the model is still trained by continuous score matching [15], and the reverse process is simulated as a geodesic random walk. Projecting an ambient diffusion onto a submanifold, however, leaves the forward transition kernel $p(\mathbf{x}_t|\mathbf{x}_0)$ without a closed form [40]: it must be simulated step by step, which removes the ability to jump to an arbitrary timestep t, making training prohibitively expensive.

3.3 The Diffusion Formulation of DiffDock

To overcome the inefficiency of numerical forward sampling, DiffDock diverges from the ambient-projection method: Corso et al. [6] define a one-to-one mapping from the molecular pose space onto a geometrically simpler manifold on which the diffusion kernel admits a closed form. This restores the central advantage of diffusion training: generating noisy states and their scores analytically.

Rather than a pose $\theta \in \mathbb{R}^{3n}$ (n atoms, m rotatable bonds), DiffDock treats the pose as an element of an $(m+6)$-dimensional submanifold $\mathcal{M} \subset \mathbb{R}^{3n}$: six rigid-body degrees of freedom from rototranslation relative to the protein, and m from the rotatable bonds. This factorises as a product manifold,

$$\mathcal{M} = \mathbb{R}^3 \times SO(3) \times SO(2)^m, \tag{3.10}$$

with tangent space $T_{\mathbf{x}}\mathcal{M} \cong T_{\mathbf{x}_{tr}}\mathbb{R}^3 \oplus T_{\mathbf{x}_{rot}}SO(3) \oplus T_{\mathbf{x}_{tor}}SO(2)^m$. Because the forward process acts independently on each factor, the transition kernel factorises, and the score decouples into a sum of component scores, each with a closed-form kernel. The normal distribution on $\mathbb{R}^3$, the $IGSO(3)$ distribution on $SO(3)$ [41], and a wrapped normal on $SO(2)^m$ [42]. This diffusion structure, architecture details and sampling process are visualised in Fig. A.1.

3.4 The Diffusion Formulation of SigmaDock

While torsional diffusion models such as DiffDock pioneered generative docking, Prat et al. [7] argue that their parameterisation (3.10) is geometrically entangled: the nonlinear, nonlocal map from torsional updates to Cartesian coordinates strips the induced measure of its product structure. Through this "lever effect," small angular perturbations at a ligand's core produce large peripheral displacements that couple distant degrees of freedom [42], yielding an ill-conditioned objective and stiff reverse-time dynamics prone to steric clashes [10], which we examine in Sect. 5.1.

To resolve this, SigmaDock decomposes the ligand into an irreducible set of m rigid-body fragments $\{\mathcal{G}_{F_i}\}_{i=1}^{m}$, fixing their internal geometries via holonomic constraints, so the generative task becomes predicting independent rigid transformations rather than interdependent torsional angles. The diffusion then operates over a cleanly decoupled product manifold $\mathcal{Z} = SE(3)^m$; the fragment state $z = (p, R) \in SE(3)^m$ collects translations p and rotations R.

From the generic diffusion SDE (3.4), the forward process applies independent roto-translations to each fragment,

$$dZ^{(t)} = \left[-\tfrac{1}{2}p^{(t)}, 0\right] dt + \left[dB^{(t)}_{\mathbb{R}^{m\times 3}}, dB^{(t)}_{SO(3)^m}\right], \tag{3.11}$$

with $B^{(t)}_{\mathbb{R}^{m\times 3}}$ and $B^{(t)}_{SO(3)^m}$ Brownian motion on the translation and rotation groups. To approximate the intractable score $\nabla_z \log p_t(z|\mathcal{G}_{dock})$, SigmaDock trains an $SO(3)$-equivariant network s_θ with a Riemannian score-matching objective,

$$\mathcal{L}(\theta) = \mathbb{E}\left[\left\| s_\theta(Z^{(t)}, t, \mathcal{G}_{dock}) - \nabla_z \log p_{t|0}(Z^{(t)}|Z^{(0)}) \right\|^2_{SE(3)^m}\right]. \tag{3.12}$$

Structural inductive biases, through fragment merging and soft triangulation constraints across fragment boundaries, confine generation to physically permissible manifolds, avoiding the post-hoc energy minimisation [10, 26, 25] or auxiliary confidence models [24] that other models require for chemical validity. This entangled-to-decoupled transition is the central test case for our calibration analysis: we hypothesise that SigmaDock's better-conditioned parameterisation yields a more calibrated conditional $p(\theta \mid x)$ than DiffDock's. Again, the diffusion process, architectural details and sampling are shown in Fig. A.2.

3.5 MIRA Scores and TARP curves

To evaluate the conditional distribution learned by a generative docking model without a tractable likelihood, we use two sample-based diagnostics, TARP and MIRA. This section states only the components our evaluation relies on. The docking-specific adaptations, such as the symmetry-corrected metric and the multi-reference extension, are deferred to Sect. 4.2.

3.5.1 TARP Curve

The Test of Accuracy with Random Points (TARP) assesses whether a generative posterior estimator represents uncertainty faithfully [11], with $\theta \in \Theta$ a sampled pose, θ^* the crystallised complex, and x the conditioning protein–ligand graph. An estimator $\hat{p}$ is *accurate* [11, Def. 1] iff they assign equal probability mass $\hat{p}(\theta \mid x) = p(\theta \mid x)$ for all $(x, \theta) \sim p(x, \theta)$. For a credible-region generator G (a map that returns a region $G(\hat{p}, \alpha, x)) \subseteq \Theta$ in pose space) at level $1 - \alpha$, the expected coverage probability (ECP) is the frequency with which the truth falls inside the region [11, Defs. 3–4],

$$\mathrm{ECP}(\hat{p}, \alpha, G) = \mathbb{E}_{p(x)}\, \mathbb{E}_{p(\theta|x)} \left[\mathbb{1}(\theta \in G(\hat{p}, \alpha, x)) \right], \tag{3.13}$$

and an accurate estimator yields $\mathrm{ECP} = 1 - \alpha$ for every α, tracing the diagonal of a coverage plot. The decisive ingredient is a *positionable* generator [11, Def. 5], anchored at a freely chosen reference point c to which it contracts as $\alpha \to 1$. Their main result [11, Thm. 3] states that correct coverage for such a generator, for all α, all x, and all reference points c, is both necessary *and* sufficient for accuracy $\hat{p}(\theta \mid x) = p(\theta \mid x)$ (given shared support under continuity). In practice, TARP uses spherical regions [11, Def. 7], reducing the test to a counting rule: for each complex one draws a reference point c and computes the fraction of posterior samples nearer to c than the ground truth θ^*. The truth lies in the $1 - \alpha$ region iff that fraction over all complexes is $\leq 1 - \alpha$ (visually explained in Fig. 3.1). Because the test is insensitive to the reference-point distribution and to the choice of L_1 or L_2 distance [11], it transfers cleanly to docking. Our multi-reference extension and the symmetry-corrected metric are described in Sect. 4.2. The resulting TARP curve, averaged over the test set, is shown in Fig. 5.3.

3.5.2 MIRA Score

Complementing TARP, the MIRA score quantifies the alignment between a candidate conditional and the true data-generating process [12], in the same single-ground-truth regime (one crystal pose per complex) and without Bayesian evidence computations. Using the principle that two distributions coincide iff they assign equal mass to every region, MIRA builds a random region around a sampled centre [12, Eq. 3], $R = \{\theta : d(\theta, c) \leq d(\theta_r, c)\}$, where the reference pose θ_r is drawn from the candidate $p(\theta \mid x_i)$ and the centre c from an arbitrary $p(c)$. It records $n = \sum_j \mathbb{1}(\theta_j \in R)$, the candidate samples inside, and $k = \mathbb{1}(\theta^* \in R)$, whether the single true pose lies inside [12, Eq. 4], with $n \sim B(N, \lambda_n)$ and $k \sim B(1, \lambda_k)$ (visually explained in Fig. 3.2).

Because $\lambda_n \mid x^*, c$ is uniform on $[0, 1]$ [12, Prop. 3.1] derived by a probability integral transform of

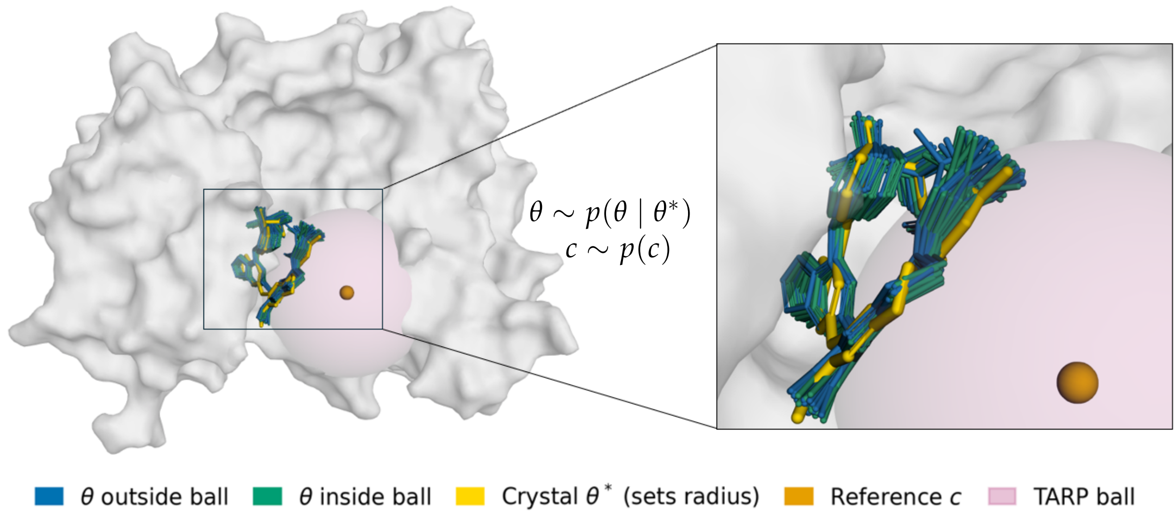

Figure 3.1: Visualisation of the TARP evaluation method applied to a generated pose ensemble. A reference point c (orange sphere) is randomly sampled to anchor a spherical credible region (pink). The radius of this region is strictly defined by the distance to the experimental ground-truth crystal pose θ^* (yellow). In this specific scenario, out of $N = 40$ total generated poses, exactly 20 poses fall inside the boundary (green, $d(\theta, c) < d(\theta^*, c)$) and 20 poses fall outside (blue). This spatial distribution yields a fractional mass of $f_{ik} = 0.5$, demonstrating that the crystal ground truth sits precisely at the 50th percentile of the model's predicted conditional distribution relative to this chosen reference point.

$D = d(\theta, c)$, MIRA admits a closed-form statistic via Laplace's rule of succession [12, Prop. 3.2],

$$p(k \mid n) = \left[\frac{n+1}{N+2}\right] \mathbb{1}(k=1) + \left[\frac{N-n+1}{N+2}\right] \mathbb{1}(k=0), \tag{3.14}$$

which, averaged over regions and complexes, gives the MIRA score [12, Eq. 11]

$$\mu_{\text{Mira}}(\hat{p}) = \mathbb{E}[p(k \mid n)] \approx \frac{1}{L \cdot L_r} \sum_{l,r} p(k_{l,r} \mid n_{l,r}). \tag{3.15}$$

Under the null that the candidate distribution and truth coincide, $p(k \mid n)$ converges to $\text{Beta}(2,1)$ with mean $2/3$ and variance $1/18$. The finite-sample null $\mu = (2N+3)/(3(N+2))$ with band $2/3 \pm \sqrt{1/(18L)}$, is used throughout. The score is bounded below by $1/2$ (disjoint support) and asymptotically $\mu \to 1/2 + 2\,\text{Cov}(\lambda_n, \lambda_k)$, so a value below $2/3$ signals overconfidence and above it underconfidence. Two caveats matter: the $2/3$ criterion is necessary but not sufficient [12, §3.3], and, mirroring TARP, an uninformative estimator $\hat{p}(\theta \mid x) = p(\theta)$ can falsely score exactly $2/3$ unless the centres depend on x. We return to this in Sect. 4.6.5, when adapting the methodology to the molecular docking space. Similar to TARP, their sensitivity study makes it suitable for our setting.

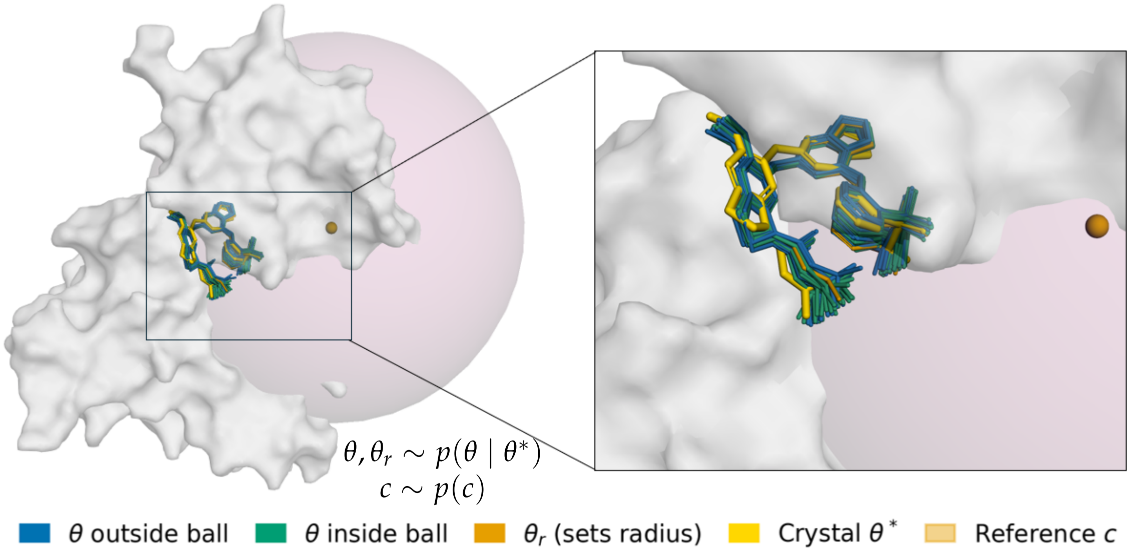

Figure 3.2: Visualisation of the MIRA evaluation method. An arbitrary centre c (light orange) is sampled to anchor a spherical test region (pink). Crucially, unlike the TARP method, the radius of this region is established by the distance to a reference pose θ_r (orange) drawn directly from the model's generated candidate distribution, rather than the ground truth. To evaluate the posterior, two values are recorded: n, the total number of candidate poses falling inside the boundary (green, $d(\theta, c) \leq d(\theta_r, c)$), and k, a boolean indicator of whether the true experimental crystal pose θ^* (yellow) is successfully captured within this specific region. These counts are subsequently aggregated to compute the sample-based MIRA score via Laplace's rule of succession $p(k \mid n)$. In this specific case of $k = 0$, $n = 23$, the formula yields a MIRA score $p(k \mid n) = \frac{40-23+1}{40+2} = \frac{16}{42} \approx 0.38$, which would need to be averaged over a large number of random centres and complexes.

4 Methods

In this section, we provide an overview of the utilised benchmarks and metrics, delve into the architectural details of DiffDock and SigmaDock, and adapt the TARP/MIRA frameworks.

4.1 Data and Benchmarks

For reliable accuracy and generalizability measurements, it is important to mitigate any train-test set leakage and have a representative test set. We evaluate the models on two benchmarks. Originally, DiffDock adopts a time-split of PDBBind v2020 [43]: 363 protein–ligand complexes discovered (crystallised) in 2019 or later, with no ligand overlap with the training set, a more realistic setting than earlier random splits [2, 6]. The PoseBusters benchmark [10] follows this and is created with 308 PDB complexes released after 2021.

1. **PDBBind** Only the PDB-IDs of the test set were available. Of the 363 complexes, 41 (11%) were excluded because their ligand SDF files were missing or corrupted leaving 322 complexes for evaluation. The filetype conversion (mol2-to-SDF) failed while the protein processed successfully.

2. **Posebusters** The PoseBusters set was used preprocessed.[1] Two versions exist; v1 contains 120 more erroneous complexes than v2, and DiffDock-L was likely tested on v1 [24]. We evaluate on the 305/308 complexes that processed successfully.

For reproducability, we publish our inference results in this drive [2].

4.2 Metrics

A generated pose is a successful docking if it deviates little from the crystal structure, conventionally within 2 Å [6]. We report the symmetry-corrected heavy-atom RMSD between sampled pose and ground-truth crystal structure, which accounts for equivalent-atom symmetries such as aromatic rings [44]. Accuracy is the percentage of complexes (e.g. the top-1-ranked pose) below the threshold, rather than the mean RMSD of the generated poses [6].

4.3 DiffDock's Architecture

DiffDock is a two-stage system: a generative score model produces poses, and a separate confidence model ranks them. The score model runs the reverse diffusion on a coarse-grained protein (backbone α-carbon atoms initialised with ESM-2 protein-language-model embeddings [45]) paired with the full ligand. It builds geometric graphs of ligand atoms and protein residues, encoding chemical features such as hybridisation and charge, with a sparse, time-dependent cutoff connectivity that Corso et al. [6] report improves the inductive bias.

Across all layers the model performs $SE(3)$-equivariant message passing based on Tensor Field Networks [46], combining the scalar chemical features of the nodes with the spherical harmonics

[1] The PoseBusters test set on Zenodo.
[2] All inference outputs

of the spatial edge vectors so that the internal representations intrinsically encode directional geometry. The spherical harmonics' $SE(3)$-equivariance makes them well suited to molecular docking. After several such layers, the pooled atomic representations map into the tangent spaces of the product manifold (3.10), yielding the translational, rotational, and torsional score components. The confidence model then scores the generated poses at all-atom resolution to select the top-ranked conformation. Its role is examined in the physical validity analysis of Sect. 5.1. The complete architecture is summarised in Fig. A.1.

4.3.1 The DiffDock-L Extension

We use DiffDock-L [24], which improves PDBBind accuracy by roughly 5% over the original model's 38%. The gain stems from a "Confidence Bootstrapping" self-training loop, in which the confidence model's scores act as a reward signal fed back into early diffusion steps to fine-tune the score model. Additionally, they employ substantial model scaling, enlarged training data, and a synthetic-data technique that docks extracted protein side chains as ligands. The authors attribute the bootstrapping's generalisation benefit to evaluating a pose being easier than generating one.

4.4 SigmaDock's Architecture

The SigmaDock diffusion formulation is developed in Sect. 3.4; here we summarise its score network and ranking heuristic with a visualisation in Fig. A.2. The score model $s_\theta(z, t, \mathcal{G}_{dock})$ is an $SE(3)$-equivariant graph transformer (a modified EquiformerV2 [47]) with extra triangulation edges across fragments that reinforce the soft geometric constraints of Sect. 3.4. It outputs a force vector on every atom, which is combined per fragment by summing forces into a net translation and torques into a rotation. As established in Sect. 3.4, SigmaDock uses the net force to predict an independent rigid motion per fragment rather than DiffDock's entangled global-motion-plus-torsions. The receptor is held in its crystal conformation, and the binding pocket, which supplies the canonical centre of mass, is provided as input.

Unlike DiffDock, SigmaDock has no separately trained confidence model but rather ranks the N_{seeds} generated poses with a cheap heuristic combining the Vinardo binding energy [48] with a set of PoseBusters stereochemical checks.[3] Because the heuristic is model-agnostic, we apply it directly to DiffDock's poses in Sect. 5.1 to isolate the ranker from the generative architecture.

[3]The mixed score is $s_i = -b_i / p_i^\beta$ with $\beta = 4$, where b_i is the binding energy and $p_i \in [0, 1]$, the mean validity of checks on bond lengths and angles, tetrahedral chirality, internal steric clash, and minimum distance to the protein. Higher s_i denotes a more confident pose.

4.5 Sampling Solvers of DiffDock and SigmaDock

During inference, both models discretise the reverse continuous-time process using a 1st-order Euler solver, where the cumulative discretisation error scales linearly with the step size Δt. However, the models diverge in their specific sampling heuristics:

- **DiffDock:** The default SDE sampler with 20 integration steps is employed. To modulate generative diversity, the default temperature-scaling parameters ($\tau > 1$ for rototranslation and torsion) are applied, thereby amplifying both the score-based drift and the injected Gaussian noise at each step.
- **SigmaDock:** SigmaDock's default inference configuration is a deterministic probability-flow ODE sampler with 25 integration steps. The sampler exposes a stochasticity parameter $\eta \in [0, 1]$ that scales the Gaussian noise injected for SDE sampling. At the production default $\eta = 0$ this stochastic term vanishes and the sampler reduces to the pure ODE.

SigmaDock's default deterministic sampler ($\eta = 0$) deliberately deviates from the theoretical probability-flow ODE (3.6): it retains the full $g(t)^2$ score-drift coefficient rather than the $\frac{1}{2}g(t)^2$ that preserves the marginal densities, dropping the compensating stochastic term:

$$d\mathbf{x} = \left[\mathbf{f}(\mathbf{x}, t) - g(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x})\right] dt \tag{4.1}$$

Omitting the $\frac{1}{2}$ factor pushes predictions twice as hard toward high-score modes, which has a mode-seeking bias [16] whose effect on coverage we quantify with the MIRA statistic. This configuration is used as the SigmaDock default in all subsequent baseline comparisons.

Employing the TARP and MIRA frameworks of Sect. 3.5, we quantitatively assess the calibration of both models under the whole-pose symmetry-corrected RMSD of Sect. 4.2. Because the score model supports both numerical SDE solvers and probability-flow ODEs, we further investigate how these integration schemes and the number of discretisation steps affect the fidelity of the learned posterior.

4.6 TARP and MIRA Validity and Adjustments

Following the TARP and MIRA foundation, we established in Sect. 3.5, we now move on to the special case of molecular docking and per diffusion manifold decomposition. The corresponding docking implementations are made public on this repository [4].

4.6.1 Multi-Reference Extension

This work extends the TARP estimator by drawing K reference points per complex rather than one. Each f_{ik} stays marginally Uniform$[0, 1]$ under a calibrated estimator regardless of K, so the pooled ECP remains an unbiased estimate of the diagonal. $K > 1$ simply densifies the expectation over reference points and smooths the empirical curve at the finite sample sizes of our benchmarks:

[4] Code on GitHub

$$f_{ik} = \frac{1}{N} \sum_{j} \mathbb{1}[d(\theta_{ij}, c_{ik}) < d(\theta_i^*, c_{ik})], \quad k = 1, \ldots, K \tag{4.2}$$

$$\mathrm{ECP}(\alpha) = \frac{1}{L \cdot K} \sum_{i=1}^{L} \sum_{k=1}^{K} \mathbb{1}(f_{ik} \leq 1 - \alpha), \tag{4.3}$$

with L complexes and N samples per complex. We use the symmetry-corrected RMSD ($\approx \mathbb{R}^{75}$) in our experiments, while illustrating the test on the molecule's centroid ($\mathbb{R}^3$) in Fig. 3.1.

4.6.2 Requirements on Distance Function

Both tests impose only light demands on the distance d, which is what licenses the manifold-specific metric we use [11, 12]. MIRA needs only that $D = d(\theta, c)$ have a continuous CDF under the candidate, from which its probability integral transform follows. TARP needs the identity of indiscernibles together with a radius monotone in α, so that its credible balls are positionable. Crucially, neither test requires d to be symmetric or to satisfy the triangle inequality, so a non-metric symmetry corrected RMSD is admissible.

4.6.3 Whole-Pose Metrics for Docking

As discussed in Sect. 3.3, the docking pose lives on the product manifold $\mathbb{R}^3 \times \mathrm{SO}(3) \times \mathrm{SO}(2)^m$ (Eq. 3.10) [6]. We take the whole-pose test as the joint calibration verdict and utilise the symmetry-corrected RMSD already used for accuracy (Sect. 4.2):

$$\mathrm{symRMSD}(\theta, c) = \min_{\sigma \in \mathrm{Sym}} \mathrm{RMSD}(\sigma\theta, c) \tag{4.4}$$

is computed with spyrmsd [44]. As a continuous metric on the quotient space $\mathbb{R}^{3N}/\mathrm{Sym}$, it satisfies both requirements above, so inference is correctly interpreted modulo equivalent-atom relabelling. All distances are taken in the receptor frame without rigid alignment, so translation, rotation, and torsion are retained jointly rather than partly removed by superposition.

Unless otherwise stated, we will use the following parameters in all experiments. MIRA and TARP have been proven to work in this regime of dimensions, fiducials, and samples ([12, 11]).

- $N = 40$ samples per complex
- $L = 322/305$ for PDBBind/Posebusters test set
- $L_r = 100$ number of regions in MIRA
- $K = 10$ number of centers c_k
- $\mathbb{R}^{3m}$-dimensional space, m(ligand) number of dimensions, median$(m) \approx 25$ ($\approx \mathbb{R}^{75}$)

Uncertainty on all ECP curves and MIRA scores comes from a 500-replicate nonparametric bootstrap over complexes, block-resampling each complex's K TARP fractions together to respect within-complex correlation. We report the 5th–95th percentile band for ECP curves and

± 1 standard deviation for MIRA, following Sharief et al. [12].

4.6.4 Per-Group Decomposition

Alongside the joint test, we evaluate three marginal tests, each with a manifold-appropriate metric. Translation uses the Euclidean distance between centroids:

$$d_t(\theta, c) = \|t_\theta - t_c\|_2 \text{ on } \mathbb{R}^3 \tag{4.5}$$

Because the centroid is a permutation-invariant, no symmetry correction is needed. Rotation uses the SO(3) geodesic angle:

$$d_R(R_y, R_c) = \arccos((\mathrm{tr}(R_\theta^\top R_c) - 1)/2) \tag{4.6}$$

Torsion uses the root-mean-square wrapped angular difference:

$$d_\tau(\varphi_\theta, \varphi_c) = \sqrt{(1/m) \sum_b \Delta_{\mathrm{wrap}}(\varphi_b^\theta, \varphi_b^c)^2} \tag{4.7}$$

with $\Delta_{\mathrm{wrap}}(a, b) = \mathrm{atan2}(\sin(a-b), \cos(a-b)) \in (-\pi, \pi]$, so that $179°$ and $-179°$ differ by $2°$ rather than $358°$, evaluated under the spyrmsd atom permutation.

Each group distance depends on a single pose component, so its region is a cylinder rather than a ball and is not positionable in the joint space. TARP's sufficiency (see Theorem 3) therefore does not apply, making these necessary-only marginal tests. Passing all three need not imply joint calibration, since miscalibration in the correlation between components, for example, translation and rotation, is invisible to axis-wise tests. We thus report the group scores only as diagnostics for localising error, retaining the whole-pose symRMSD as the verdict on joint calibration.

4.6.5 Prior of Reference Points

One failure mode must be guarded against: an uninformative estimator $\hat{p}(\theta \mid x) = p(\theta)$ may return perfect coverage ECP $= 1 - \alpha$ and is invisible to the test unless the reference distribution $\tilde{p}(\theta_r \mid x)$ is made to depend on x [11, App. D, §4.3]. Each reference is a random pose of the true ligand $x = (P, L)$, placed relative to its receptor, as in DiffDock's initial diffusion distribution. The true pose is corrupted by a normal distribution for translation, Haar on SO(3) for rotation, uniform for torsion [6]. Since the RMSD metric uses the ligand graph and translation is centred on the receptor, the reference distribution depends on x. However, when testing coverage within a single diffusion group (rotation or torsion alone), the reference is x-independent, and TARP and MIRA lose this guarantee.

5 Results and Discussion

This thesis confirms that both models, DiffDock and SigmaDock, are reproducible under default inference with 40 poses per complex, a prerequisite for a realiable trustworthiness analysis that follows. Table 5.1 compares our reproduced accuracies against the published values. All deviate by at most 5.8% and the discrepancies are fully attributable to documented dataset differences (DiffDock-L was likely benchmarked on PoseBusters v1, which contains 120 additional erroneous complexes, and our PDBBind set was reduced to 322/363 complexes by SDF-conversion failures, Sect. 4.1). The physical validity of DiffDock-L (27%) has, to our knowledge, never been published, but is consistent with the 12% reported for the original DiffDock [10]. For SigmaDock the test set is identical, so the small residual gaps (+0.6% accuracy, −2.0% validity) reflect sampler stochasticity across seeds and version differences in the Vinardo [48] ranking rather than the data.

Model	**Benchmark**	**RMSD < 2 Å**		**+ PB valid**	
		Ours	Reported	Ours	Reported
DiffDock-L	PoseBusters v2	52.8%	50.0%[a]	27%	–[b]
DiffDock-L	PDBBind[c]	40.5%	43.0%	–	–
SigmaDock	PoseBusters v2	81.1%	80.5%	77.9%	79.9%

Table 5.1: Reproduction of reported docking accuracy under default inference (40 poses per complex). All values deviate by $\leq$ 5.8% from the literature, with discrepancies traceable to dataset variation. [a]Reported on PoseBusters v1, which includes 120 additional erroneous complexes [24]. [b]The PB-filtered validity of DiffDock-L is unpublished but comparable to 12% for the original DiffDock [10]. [c]Reduced to 322/363 complexes after ligand SDF-conversion failures. SigmaDock uses its heuristic ranker (five PoseBusters checks + Vinardo affinity, see Sect. 4.4) [7, 48].

5.1 Posebuster Filtering Effect

Deep-learning docking models frequently generate physically invalid poses, and post-hoc energy minimisation corrects this only partially [10]. Because DiffDock decouples generation (score model) from ranking (confidence model, Sect. 4.3), physical validity could in principle be a property of either stage. This tests the first axis of trustworthiness, physical validity, against our central determinant: whether it is fixed by what the model learns or recoverable through the post hoc ranking stage. An invalid rank-1 pose can originate at either stage, so we disaggregate the two (Fig. 5.1A), which shows both failure modes are common. Because 37% of complexes yield no valid pose across all 40 samples on PoseBusters (30% on PDBBind), a hard ceiling is imposed by this generation failure, no reranker can exceed. A further 27–30% generate a valid pose that the confidence model then fails to select. The ranker discriminates validity only weakly ($\rho = +0.48$; top-5 validity $\approx$ 34% vs. $\approx$ 15% for the bottom-5), and forcing selection of the first valid pose barely improves RMSD accuracy. Generation, not ranking, is the dominant bottleneck.

The failure is overwhelmingly due to steric clashes: the minimum-distance-to-protein check fails for all but 9 of 322 complexes, and these clashing poses sit close to the crystal centroid (similar to what was reported for the original DiffDock [10]). The model therefore places the

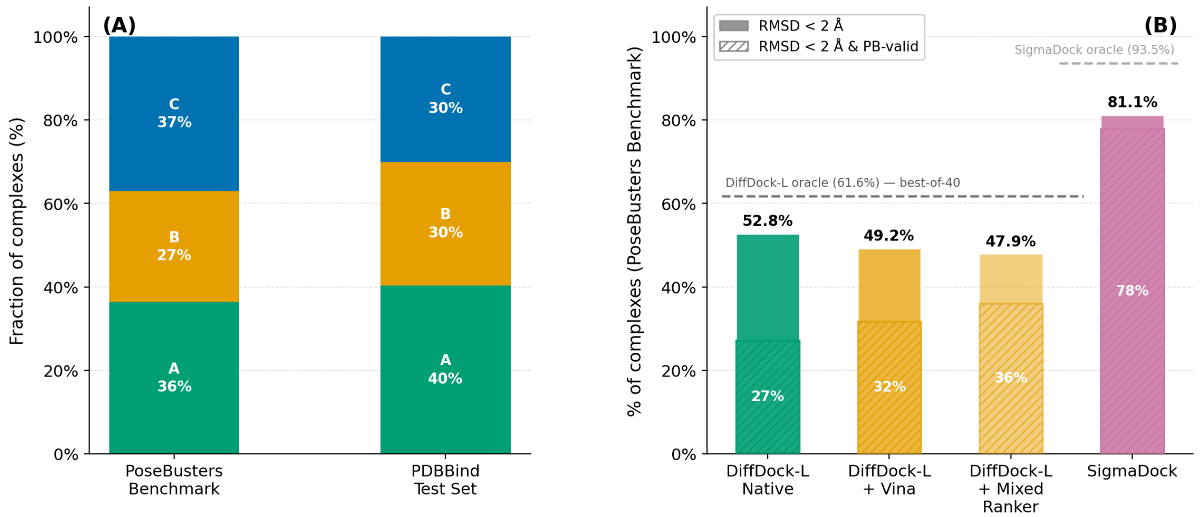

Figure 5.1: Disaggregating generative and ranking failures in molecular docking. (A) Root-cause analysis of physically invalid rank-1 DiffDock predictions. Category A: valid rank-1 pose. Category B: valid pose generated but missed by the ranker (confidence model failure). Category C: no valid pose generated (score model failure). The high frequency of Category C failures establishes a hard performance ceiling for any reranking strategy. (B) Top-1 PoseBusters validity across ranking paradigms. Replacing DiffDock's native confidence model with Vina affinity or SigmaDock's mixed heuristic (which includes five PB checks) provides only negligible improvements. The large gap between reranked DiffDock and native SigmaDock demonstrates that state-of-the-art physical validity is driven by generative inductive biases rather than post-docking ranking.

ligand in roughly the right pocket but mis-resolves its local geometry, which is precisely the lever effect anticipated for DiffDock's entangled torsional parameterisation (Sect. 3.4) [42]. An explicit steric penalty during generation is a promising remedy [23], as is the proposed architecture of SigmaDock [7]. Consistent with this generative origin, applying SigmaDock's ranker directly to DiffDock's poses raises validity only marginally (Fig. 5.1B), far short of SigmaDock's native rate. This settles the first axis of trustworthiness: physical validity is governed by the generative architecture, not the ranking stage, because post-hoc selection cannot manufacture valid geometry that the score model never generated. Whether this architectural primacy extends from individual poses to the full posterior is the question we take up next.

5.2 Coverage Test Stability

Having established that DiffDock's pointwise failures are of generative origin, we turn to the second axis of trustworthiness, whether the posterior reflects honest uncertainty. Does the model assign correct probability mass to its poses? We answer this primarily with the MIRA score, which captures the same miscalibration as the TARP ECP curves render graphically, so those are deferred to Appendix B [12, 11].

For our sample size ($N = 40$), the perfectly calibrated posterior score is ≈ 0.683 (Sect. 3.5.2). Across every evaluation, the MIRA scores fall below this baseline (Fig. 5.2), revealing pervasive overconfidence: even when the model locates the correct pocket, its samples fail to span the full volume of plausible phase space, concentrating the posterior mass too tightly. This is

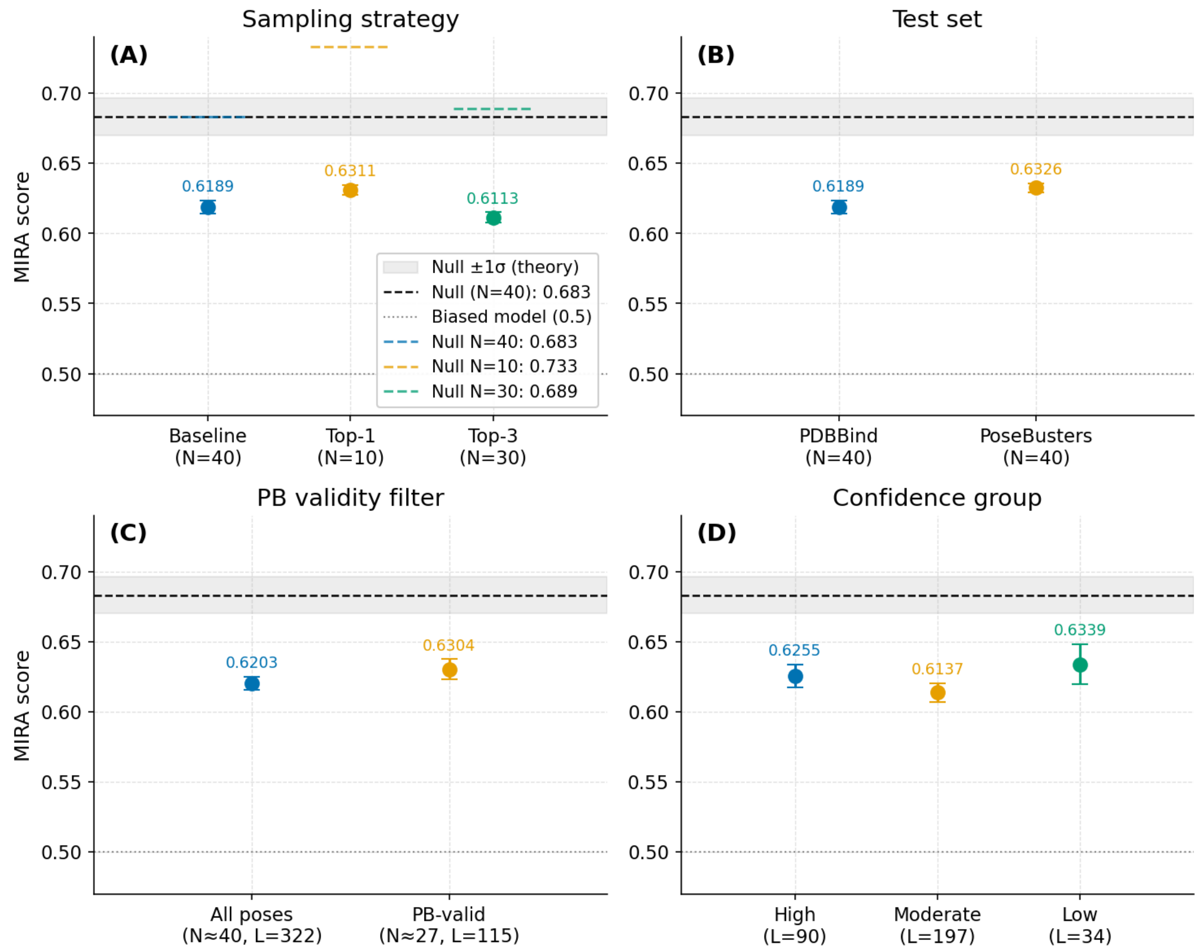

Figure 5.2: Robustness of MIRA score miscalibration across evaluation conditions. The panels display MIRA scores broken down by dataset, protein family, sampling strategy, and ligand flexibility. Dashed lines indicate the theoretical reference score for a perfectly calibrated posterior (≈ 0.683 for $N = 40$) and 0.5 for fully biased model. The grey theoretical error band uses $L = 320$ in all panels, even though some have fewer complexes (e.g. plot C with $L = 115$). Across all evaluated dimensions, the MIRA scores consistently fall below this reference threshold. This demonstrates pervasive overconfidence in DiffDock's learned posterior, confirming that the observed miscalibration is a fundamental structural property of the model rather than an artefact of the specific testset, target class, or sampling methodology.

not an artefact of the evaluation setup. The scores are nearly indistinguishable between the PDBBind and PoseBusters test sets and persist across protein families, sub-sampling strategies, and ligand-flexibility bins, identifying the miscalibration as a structural property of the learned posterior rather than of any particular benchmark or target test case.

Finally, calibration is strictly orthogonal to both pointwise accuracy and physical validity: cross-correlating against the PoseBusters checks of Sect. 5.1 shows steric invalidity and miscalibration to be parallel, independent failure modes. Together with the validity result of Sect. 5.1 this exposes three independent axes of model quality, point accuracy, physical validity, and posterior calibration, only the first of which a conventional RMSD leaderboard reports.

This settles the calibration axis in the negative: generative docking inherits the systematic overconfidence repeatedly found in simulation-based inference [9]. Next, we investigate how much of it is fixed by the architecture versus sampler variations.

5.3 Model Comparison DiffDock vs SigmaDock

This comparison is framed by two caveats. First, the models target different tasks: DiffDock performs blind docking over the entire receptor surface, whereas SigmaDock performs pocket-specified docking with a known pocket and a fixed receptor (Sect. 2.1). Second, they differ in training data: SigmaDock is trained on the PDBBind train–test split (≈19k complexes) for a fair comparison with the original DiffDock, whereas the model we evaluate is the scaled, synthetic-data-augmented DiffDock-L (Sect. 4.3.1). The two are therefore not directly rankable on accuracy; we instead read the whole-pose TARP and MIRA results as evidence on posterior calibration, the property that top-1 accuracy leaves unmeasured.

Both models are overconfident: each traces the characteristic S-curve and scores below the calibrated baseline of 0.683 (Fig. 5.3), so the miscalibration is not unique to DiffDock. The severity, however, differs sharply. SigmaDock's coverage curve tracks the diagonal more closely, with an overall MIRA deviation roughly 1.8× smaller than DiffDock's (0.655 vs. 0.633). This is the direct empirical confirmation of the hypothesis advanced in Sect. 3.4: SigmaDock's decoupled $SE(3)^m$ parameterisation yields a better-conditioned, more calibrated posterior than DiffDock's entangled product manifold. This calibration gain moves in line with SigmaDock's higher geometric accuracy and physical validity rates.

But one confound remains, because the default configurations differ not only in architecture but in sampler (SigmaDock's deterministic ODE-25 against DiffDock's stochastic SDE-20). The comparison above cannot attribute the calibration gap to the parameterisation alone rather than the integration scheme. Disentangling the two is addressed through the group-wise and sampling-strategy analyses that follow.

5.3.1 Coverage Test by Diffusion Group

To localise the calibration gap of Sect. 5.3, we decompose the posterior into its three diffusion groups (translation, rotation, and torsion), which DiffDock models on a coupled product manifold, but SigmaDock factorises into independent rigid fragments (Sect. 3.4). These marginals are necessary-only diagnostics (Sect. 4.6.4), so the whole-pose RMSD remains the main metric of choice. Because the groups are only weakly correlated in top-1 accuracy, with translation–rotation $\rho = 0.48$ and the others $\rho \leq 0.19$) and each marginal mirrors the whole-pose miscalibration, the overconfidence is intrinsic to the individual manifolds rather than inter-group coupling.

DiffDock is overconfident in every group (Fig. 5.4), most severely in translation (0.632; rotation 0.649, torsion 0.642): it is least calibrated about where to place the ligand centroid. SigmaDock improves all three, but the signature result is its near-perfect torsional calibration (0.674; translation 0.652, rotation 0.661), corroborated by the group-wise TARP curves (Appendix Fig. B.2). The architectural advantage is therefore confined. It concentrates on torsion, the exact degree of freedom, for which SigmaDock's fragment decomposition is built to decouple.

This is the mechanistic half of the architecture-versus-sampler question: the whole-pose gain

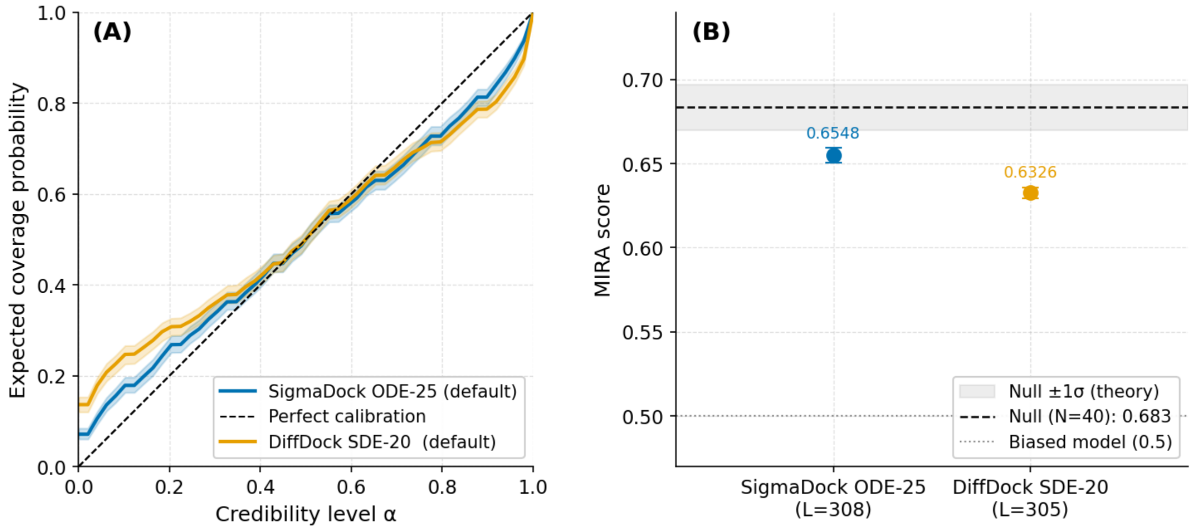

Figure 5.3: Comparative evaluation of posterior calibration between DiffDock and SigmaDock. (A) TARP Expected Coverage Probability (ECP) curves evaluated on whole-pose symmetry-corrected RMSD. The dashed diagonal represents perfect theoretical calibration. Both models show an S-curve, signifying overconfidence in their spatial predictions. However, SigmaDock's default ODE-25 sampler (blue) tracks noticeably closer to the diagonal than DiffDock's SDE-20 sampler (yellow). (B) Corresponding MIRA scores quantify the TARP results. Both models fall below the theoretical null baseline of 0.683 for $N = 40$ samples, confirming overconfidence. Yet SigmaDock achieves a significantly higher score (0.655) than DiffDock (0.633). This demonstrates that SigmaDock's architecture provides a superior quantitative alignment with the true conditional distribution and more accurately captures the underlying uncertainty of the binding manifold.

of Sect. 5.3 traces to the very manifold the inductive bias targets, strong evidence that it is architectural rather than incidental. It also closes the loop with the validity finding of Sect. 5.1: the entangled torsional updates behind DiffDock's local-geometry clashes are the same updates that miscalibrate its torsional posterior. Consequently, SigmaDock's decoupling resolves both symptoms, invalidity and overconfidence, of a single root cause.

5.3.2 Coverage Tests on Different Sampling Strategies

We separate architecture from sampler (confounded at default, Sect. 5.3) by varying both the solver and the step count for each model (Tab. 5.2, Fig. 5.5). Note that SigmaDock's default ODE is itself a mode-seeking deviation: it omits the theoretical $\frac{1}{2}$ factor (Sect. 4.5) and drives twice as hard toward high-score modes.

Matching DiffDock to an ODE-25 solver shrinks the whole-pose MIRA gap to roughly 1%, from the 2% seen at default, and renders the group-wise TARP curves of both models nearly indistinguishable (Appendix Fig. B.3). Part of the default gap was therefore the sampler, not the architecture. Yet SigmaDock retains the lead at every matched setting, and its calibration is stable across the entire sampler grid, whereas DiffDock's swings more widely. This robustness is the clearest sign that the advantage is architectural: a well-conditioned posterior is insensitive to how it is integrated.

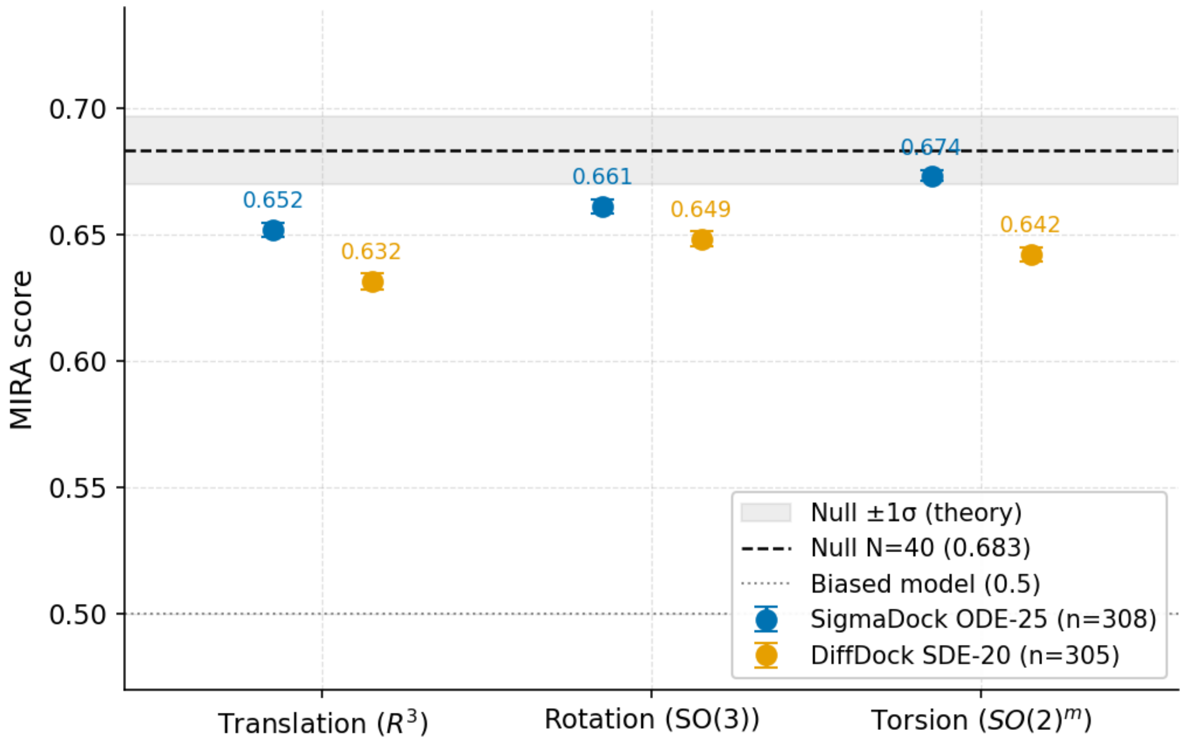

Figure 5.4: Marginal MIRA scores by diffusion group for DiffDock and SigmaDock. The scores evaluate the isolated calibration of the learned posterior across translation ($\mathbb{R}^3$), rotation ($SO(3)$), and torsion ($SO(2)^m$) manifolds on the PoseBusters test set. The dashed line indicates the theoretical baseline for a perfectly calibrated posterior at $N = 40$ samples (0.683), while the grey band denotes the $\pm 1\sigma$ theoretical uncertainty of the statistic. DiffDock (orange) exhibits systemic overconfidence across all spatial components, most severely in the translational group. Conversely, SigmaDock (blue) demonstrates substantially improved marginal coverage due to its fragment-based inductive biases, achieving near-perfect calibration within the torsional manifold.

Within SigmaDock, the sampler is nonetheless a small lever. Its MIRA score rises monotonically as the solver uses fewer steps, ODE-25 (0.653) to SDE-25 (0.661) to SDE-10 (0.675), while top-1 accuracy moves the opposite way (ODE-25 best at 81.1%, SDE-10 lowest at 78.2%). This is the precision–coverage tradeoff documented for diffusion samplers [17, 16]: the deterministic ODE is mode-seeking and narrows coverage, whereas the SDE's Langevin noise broadens it. The accuracy-optimal and calibration-optimal configurations are thus different settings along the same trade-off path.

The two architectures, however, favour opposite samplers: SigmaDock is best calibrated under the stochastic SDE-10, DiffDock under the deterministic ODE-20. Through the same error trade-off [17], the best sampler depends on which error source binds. SigmaDock's well-conditioned, decoupled manifold keeps discretisation error small, freeing the SDE's stochastic correction to broaden coverage. DiffDock's entangled reverse dynamics (Sect. 3.4) are discretisation-limited, so the lower-truncation deterministic ODE wins, while the added SDE noise merely sharpens an already-imperfect score into tighter, overconfident modes. One caveat: DiffDock's ODE run also used neutral temperature ($T{=}1$), whereas its default SDE inflates drift and noise with fitted $\tau > 1$; equalising it (SDE $T{=}1$) recovers most, though not all, of the apparent ODE advantage (Appendix Fig. B.4). Additionally, DiffDock's top-1 accuracy comes from a confidence model trained on the default reverse-diffusion sampler [6, 24], so part of DiffDock's accuracy degradation (SDE-10 collapse) plausibly reflects ranker distribution shift, not failed denoising alone.

Step count exposes the same conditioning divide. At ten steps, DiffDock fails catastrophically, with accuracy collapsing from 52.8% to 21.3% and the MIRA score overshooting the null into underconfidence (0.697), because ten steps cannot denoise its stiff dynamics. SigmaDock at SDE-10 instead reaches its best calibration at near-constant accuracy and retains the perfect torsional coverage of the per-group analysis above, while pushing to 50 steps buys only runtime. Step-size robustness is therefore itself an architectural property: the decoupled manifold tolerates aggressive step reduction, the entangled one does not.

These results resolve the architecture-versus-sampler question. Posterior fidelity is set primarily by the architecture, and secondarily by the sampler within it. For any application that consumes the posterior rather than the single top pose, SigmaDock SDE-10 is the recommended default: the best calibration and the fastest runtime, at a cost of only ≈ 2 accuracy points over the accuracy-optimal ODE-25. That the field tunes and reports these models on top-1 accuracy alone, which rewards exactly the mode-seeking deterministic samplers that suppress coverage, is precisely why this lever has gone unexamined.

Taken together with Sects. 5.1–5.3.2, the picture is complete: a model's posterior can be untrustworthy on either axis, invalid or overconfident, and both are governed first by the architecture and only then by the sampler.

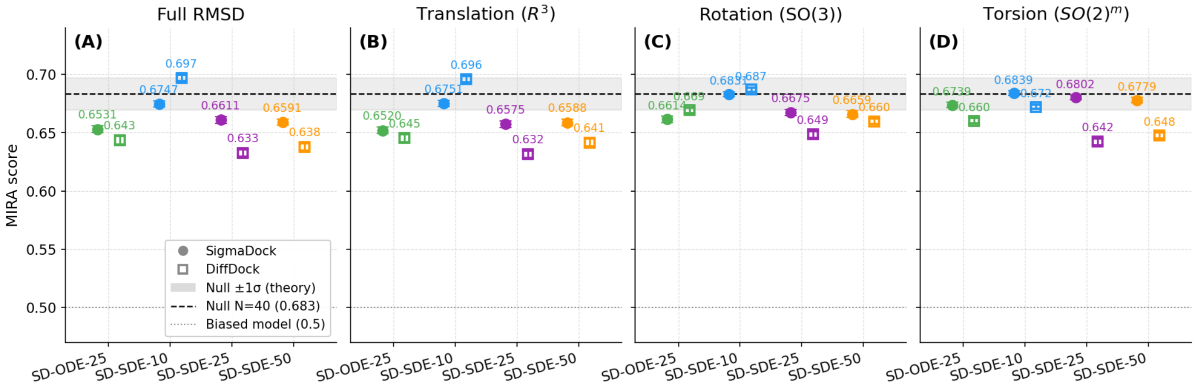

Figure 5.5: MIRA score calibration across sampling strategies and diffusion manifolds. The panels detail the posterior calibration for (A) the whole-pose full symRMSD, alongside the marginal (B) translation, (C) rotation, and (D) torsion groups. SigmaDock (circles) and DiffDock (squares) are evaluated across four inference configurations: ODE-25 (green), SDE-10 (blue), SDE-25 (purple), and SDE-50 (orange). The dashed black line denotes the theoretical baseline for perfect calibration. DiffDock exhibits severe sensitivity to discretisation errors; at 10 SDE steps, its entangled Cartesian parameterisation fails to fully denoise, leading to catastrophic overshoot and underconfidence across all manifolds. Conversely, SigmaDock demonstrates remarkable resilience to aggressive step reductions, achieving its optimal joint calibration at SDE-10 (0.6747). Crucially, (D) highlights that SigmaDock maintains near-perfect torsional coverage regardless of the chosen numerical solver or step size, empirically validating that its fragment-based inductive bias effectively resolves the geometric entanglement and overconfidence inherent in coupled torsional modelling.

Comparison Pair	Model & Condition	Temp	N	Accuracy		MIRA	Runtime per complex (s)			
				RMSD < 2 Å	+PB Valid		Wall	Inference	Vinardo+PB	GPU-s
ODE-25 vs ODE-20	SigmaDock (ODE-25)	noise=0.0	307	**81.1%**	77.9%	**0.6531**	6.5	∼1.1	∼5.4	260.0
	DiffDock (ODE-20)	all=1.0	305	49.2%	23.9%	**0.6435**	137.2	137.2	N/A	137.2
SDE-10 vs SDE-10	SigmaDock (SDE-10)	noise=1.0	307	**78.2%**	75.9%	**0.6747**	7.4	∼0.44	∼6.96	296.0
	DiffDock (SDE-10)	fitted	305	**21.3%**	6.2%	**0.6970**	48.4	48.4	N/A	48.4
SDE-25 vs SDE-20	SigmaDock (SDE-25)	noise=1.0	307	80.5%	77.9%	**0.6611**	4.5	∼1.1	∼3.4	180.0
	DiffDock (SDE-20)	fitted	305	**52.8%**	21.0%	0.6326	122.7	122.7	N/A	122.7
	DiffDock (SDE-20 $T = 1$)	all=1.0	305	52.1%	24.9%	**0.6395**	133.8	133.8	N/A	133.8
SDE-50 vs SDE-50	SigmaDock (SDE-50)	noise=1.0	307	79.5%	76.5%	0.6591	10.6	∼1.7	∼8.9	424.0
	DiffDock (SDE-50)	fitted	305	50.5%	26.6%	0.6379	336.2	336.2	N/A	336.2

Table 5.2: Performance, calibration, and runtime comparison across sampling conditions. Bold values are those referenced in the discussion. Accuracy includes standard RMSD < 2 Å and strictly valid binding poses (PB+RMSD < 2 Å). The null MIRA score baseline for $S = 40$ is 0.6833. DiffDock SDE runs utilise fitted inference temperatures (tr=1.170, rot=2.064, tor=7.044) unless $T = 1$ is specified. SigmaDock leverages batched parallelisation across GPUs (one per sample), shifting the primary temporal bottleneck to CPU-bound ranking tasks (Vinardo and PoseBusters), whereas DiffDock integrates ranking natively via a secondary confidence model on the GPU. The runtimes for DiffDock include all 40 samples and are therefore hard to compare to SigmaDock. It is recommended to run inference and ranking separately in SigmaDock, in which case, SigmaDock has superior runtime. Interestingly, there is no major difference in runtimes between ODE and SDE sampling.

6 Conclusion

Generative docking is tuned and ranked on a single number, top-1 accuracy, that is silent on whether a model's poses are physically valid and whether its posterior carries the right uncertainty. By adapting the TARP and MIRA frameworks to the docking manifold and treating the sampler as a first-class object of study, this thesis asked whether generative docking models can be trusted beyond that headline number, and what governs their reliability.

This thesis adds a third dimension to docking evaluation, posterior calibration, and its central finding is that generative docking is pervasively overconfident: DiffDock's posterior concentrates too tightly across every dataset, protein family, and sampling regime, a miscalibration strictly orthogonal to both accuracy and validity. Calibration thus joins physical validity, established earlier by the PoseBusters suite [10], as an axis of model quality that a top-1 accuracy leaderboard leaves unreported. On all three axes, the same lesson holds: quality is set primarily by the generative architecture and only secondarily by the sampler. DiffDock's invalid poses originate in its score model rather than in its ranker, and its entangled manifold miscalibrates the posterior. SigmaDock's fragment-based decoupling yields both valid geometry and a markedly better-calibrated posterior, and stays stable across solvers and step counts. The sampler remains a genuine but bounded lever, under a ten-step SDE, SigmaDock reaches its best calibration and fastest runtime at a minor accuracy cost, a configuration an accuracy-only search would never surface.

Together, these results argue for calibration-aware docking: architectures and training objectives targeting distributional coverage rather than point accuracy, sampler-aware ranking, and the MIRA score as a model-selection criterion. A concrete next step, since both models still use first-order Euler schemes, is the higher-order solvers surveyed earlier (Heun [16], DPM-Solver [29, 30]), whose lower discretisation error may let stiff parameterisations such as DiffDock's reach calibrated coverage at the step counts that currently break it. The adapted TARP and MIRA tests extend naturally to the joint and co-folding regimes now redefining the field [19].

A Architecture Details

The detailed architecture, diffusion process and sampling procedure of both models, DiffDock and SigmaDock, are shown in Fig. A.1/A.2.

B Supplementary Results

B.1 TARP Analysis

The TARP Expected Coverage Probability (ECP) curves are shown in Fig. B.1 (aggregate, with all robustness breakdowns as in Fig. 5.2), Fig. B.2 (per diffusion group in the default configuration), and Fig. B.3 (per diffusion group across sampling strategies). In all cases the curves qualitatively reproduce the MIRA scores of Figs. 5.2 and 5.5, including the underconfident signature of DiffDock SDE-10.

B.2 MIRA Analysis

DiffDock's MIRA scores across varying step counts and sampling strategies are shown in Fig. B.4, isolating the effect of temperature scaling on the ODE ($T = 1$) versus SDE ($T =$ default) calibration profiles already reported in Fig. 5.5.

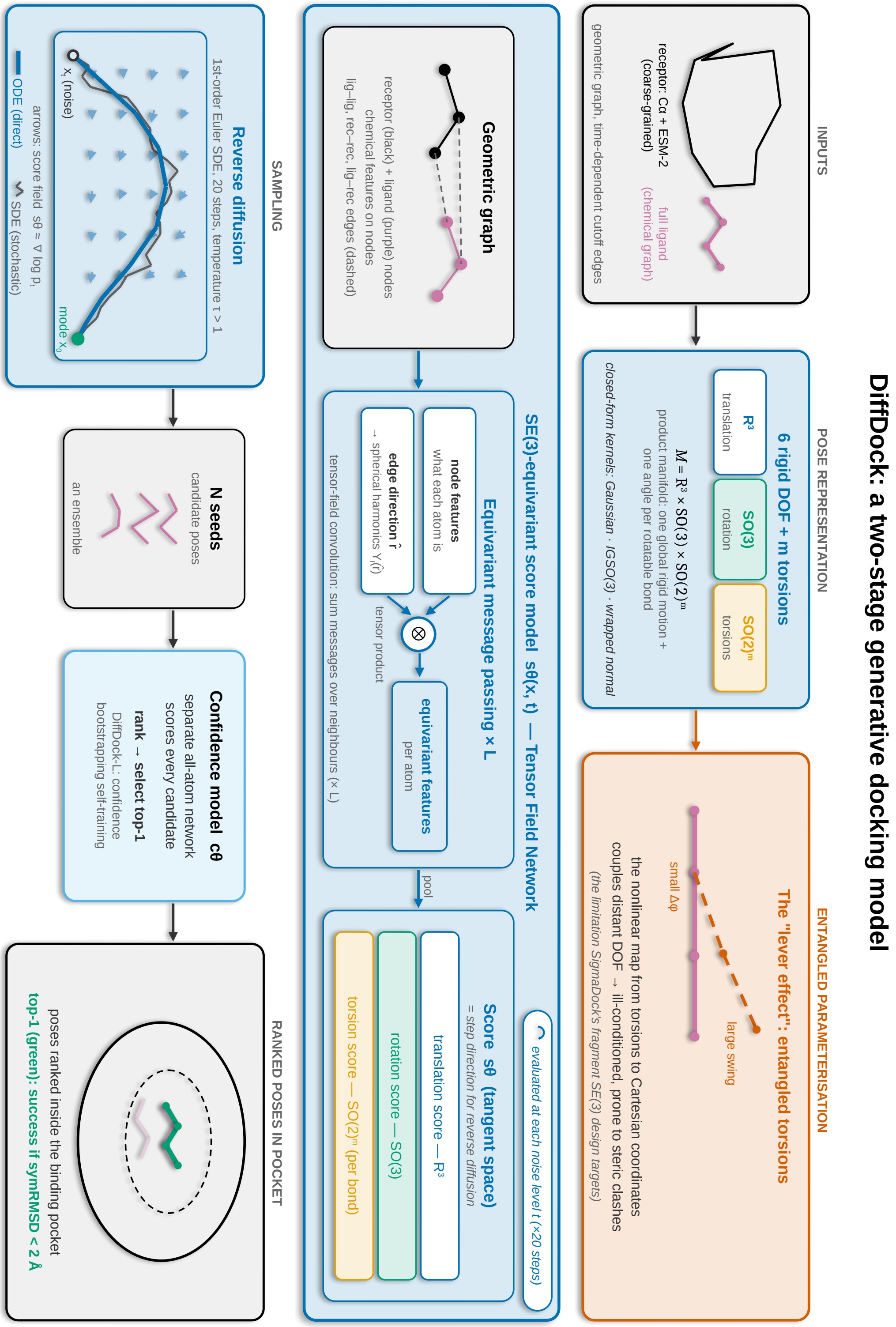

Figure A.1: Schematic overview of the DiffDock pipeline. Poses are modeled on a product manifold ($M = \mathbb{R}^3 \times SO(3) \times SO(2)^m$) coupling global rigid motion with m internal torsions. This entangled parameterisation introduces a "lever effect" in which changes in central torsion cause large peripheral displacements (Sect. 3.4). An $SE(3)$-equivariant Tensor Field Network generates translation, rotation, and torsion scores to guide reverse SDE sampling, and a separate learned confidence model ranks the final candidate poses. Sampling can be achieved through SDE or ODE solvers.

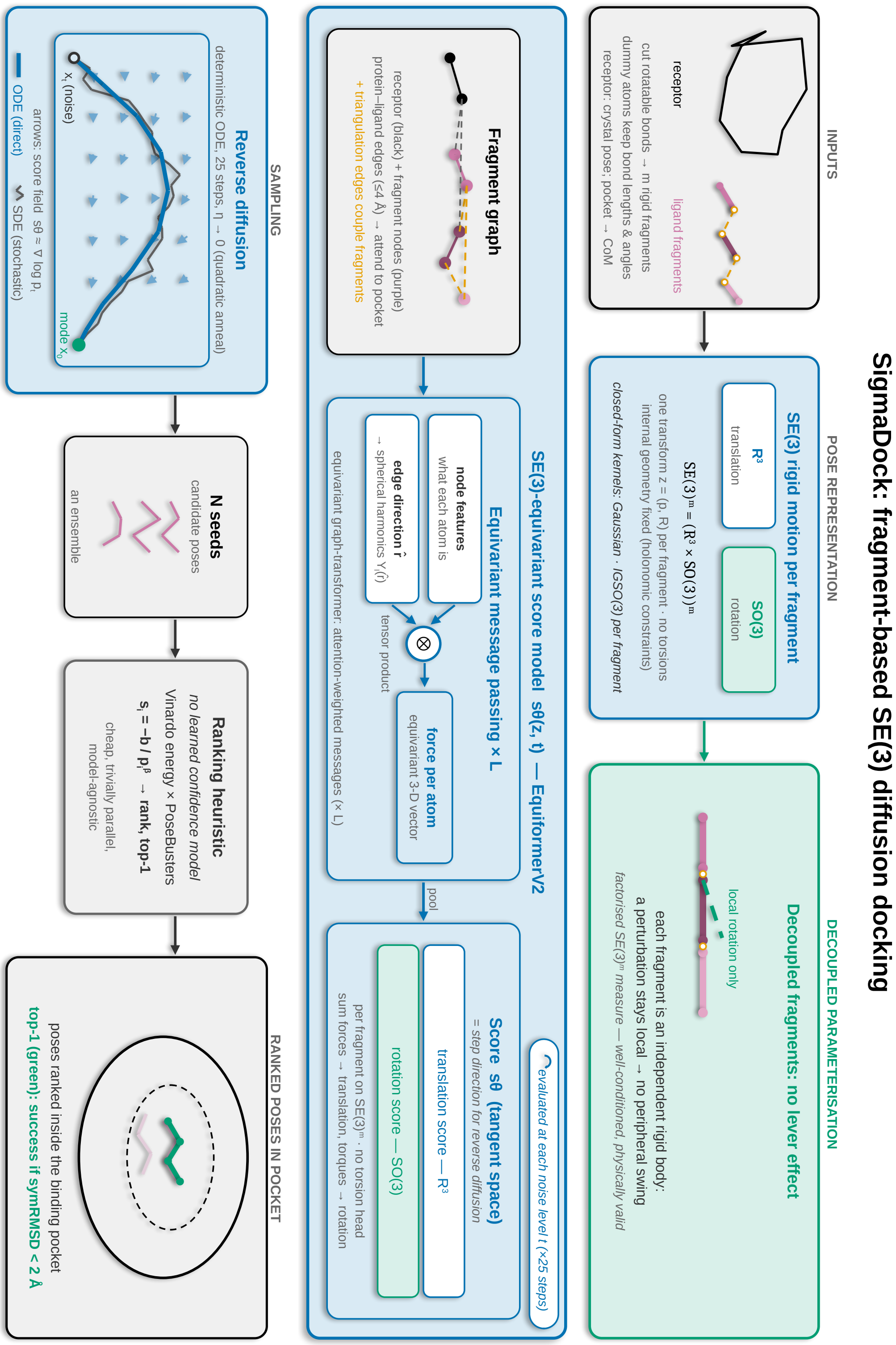

Figure A.2: Schematic overview of the SigmaDock pipeline. The ligand is partitioned into m rigid fragments across a factorised $SE(3)^m$ manifold to eliminate the lever effect and keep perturbations local. Score fields are modelled using an EquiformerV2 architecture that pools atomic forces into independent translational and rotational scores for each fragment, without a torsion head. Sampling is driven by a deterministic ODE, and poses are ranked using a model-agnostic Vinardo energy and PoseBusters heuristic.

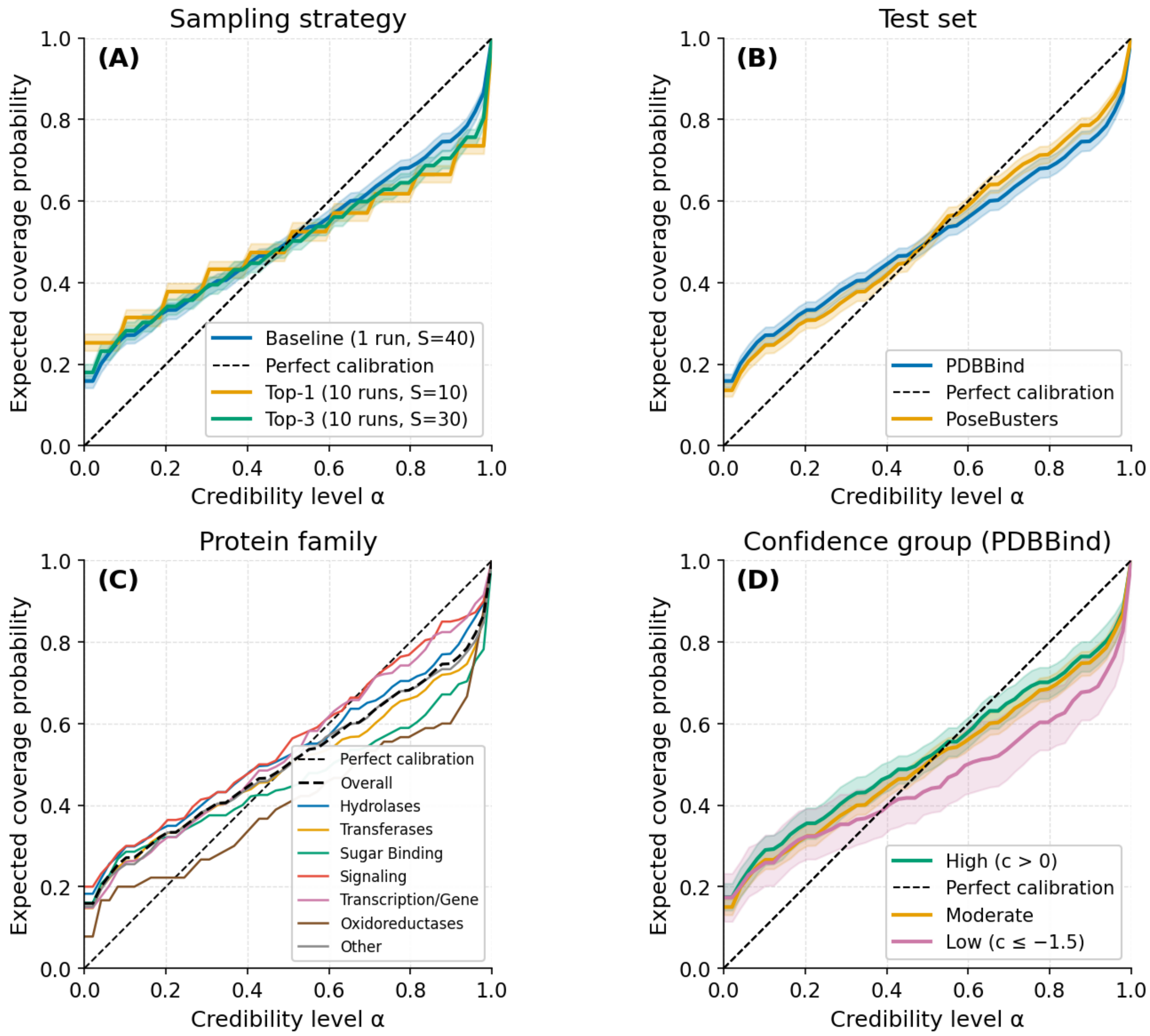

Figure B.1: TARP Expected Coverage Probability (ECP) curves. The curve is read against the theoretical diagonal: values above it indicate undercoverage and values below indicate overconfidence at a given credibility level α. The characteristic S-shape signals systemic overconfidence in the model's spatial predictions, corroborating the MIRA scores. At low credibility ($\alpha \to 0$), complete docking failures that entirely miss the binding site create a persistent 10–20% coverage floor, lifting the curve above the diagonal; at higher credibility ($\alpha > 0.3$) the curve falls below the diagonal, confirming mode collapse—the generated samples do not cover enough volume against larger reference regions. This overconfidence signature is robust and consistent across sampling strategies (A), the PDBBind and PoseBusters test sets (B), protein families (C) and confidence bins (D).

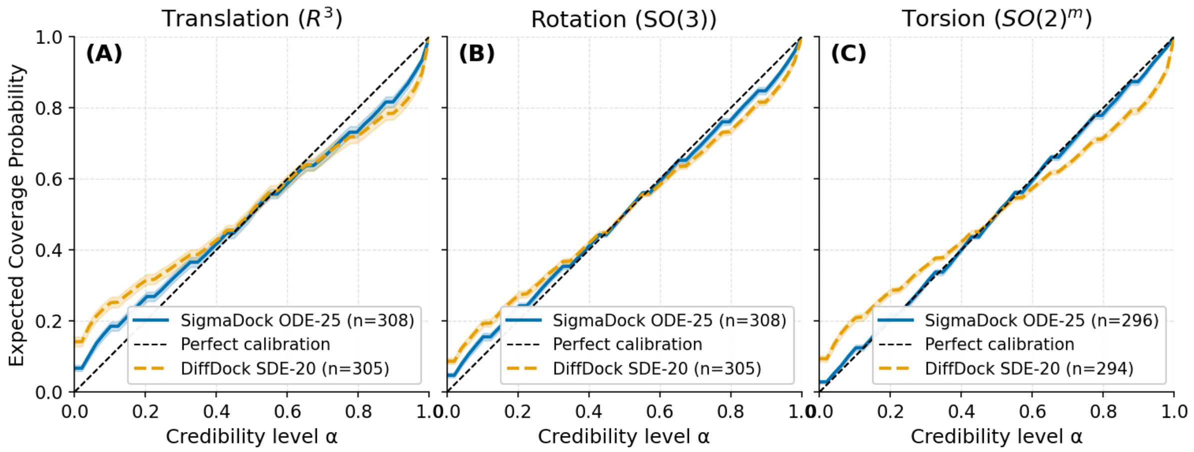

Figure B.2: Group-wise TARP Expected Coverage Probability (ECP) curves. The panels detail the marginal calibration for (A) translation, (B) rotation, and (C) torsion. The dashed black diagonal represents perfect theoretical coverage, where the empirical probability strictly matches the credibility level α. DiffDock's distributions show the same diagnostic pattern, signalling overconfidence across all degrees of freedom. SigmaDock tracks significantly closer to the ideal calibration line across all axes, corroborating the MIRA scores and demonstrating that its independent rigid-body parameterisation natively resolves much of the geometric entanglement and overconfidence seen in coupled torsional updates.

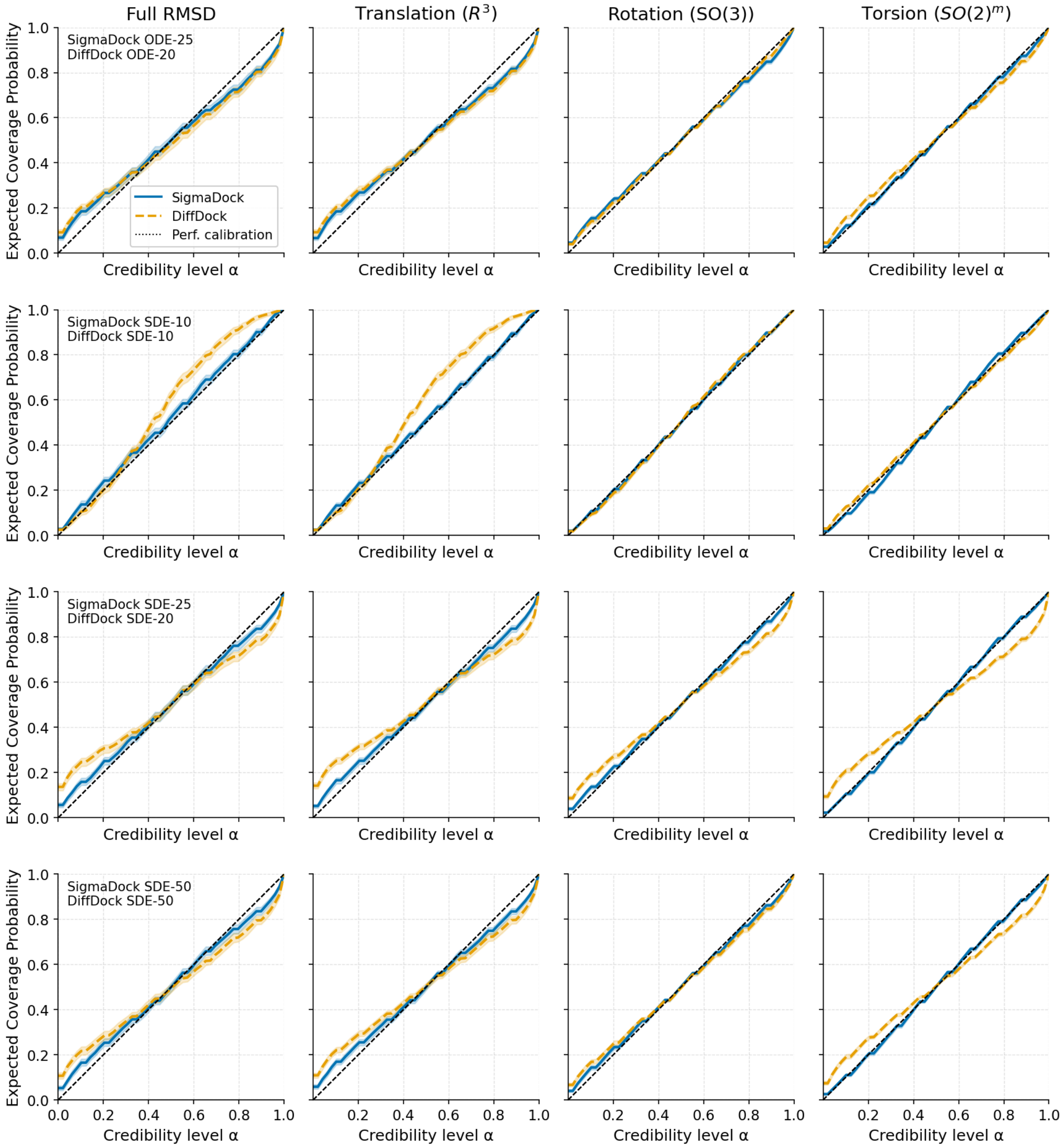

Figure B.3: Marginal TARP Expected Coverage Probability (ECP) across varied integration schemes. The matrix decomposes spatial calibration by diffusion group (columns) across paired sampling strategies (rows). The dashed diagonal represents perfect theoretical coverage. DiffDock's (yellow dashed) reliance on stochasticity and smaller step sizes is evident: at 10 SDE steps (row 2), the model fails to fully denoise the spatial representations, severely overshooting the diagonal and entering underconfidence. Conversely, SigmaDock (blue solid) demonstrates robust calibration stability across all integration schemes and step sizes. Notably, SigmaDock's independent rigid-body parameterisation natively enforces near-perfect torsional coverage (rightmost column) and remains invariant to the underlying numerical solver.

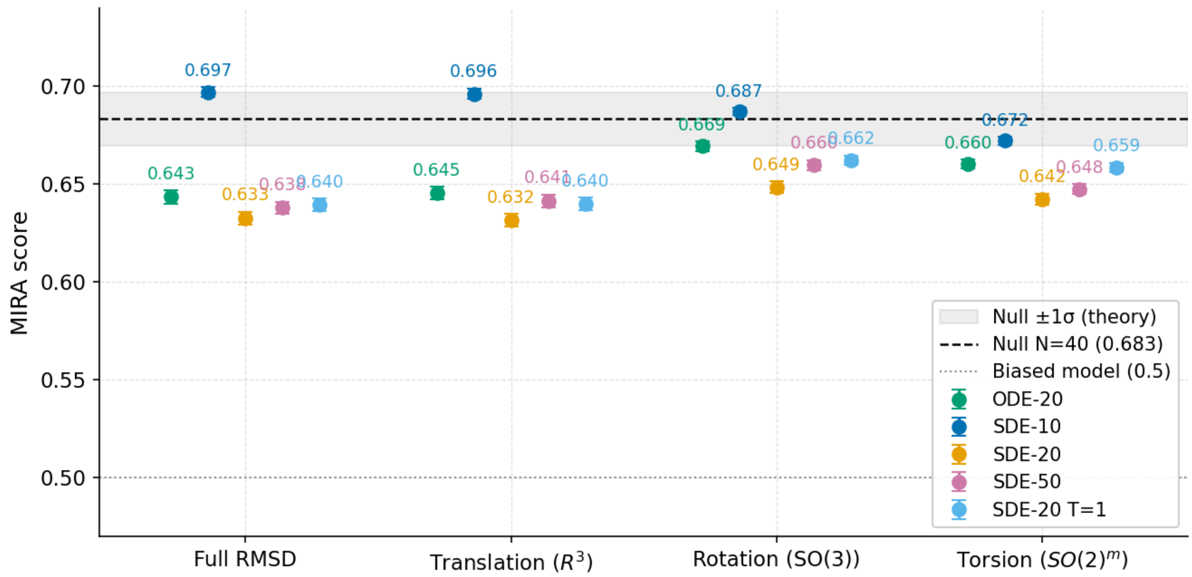

Figure B.4: Impact of discretisation and temperature scaling on DiffDock's MIRA calibration. The plot isolates the marginal and joint MIRA scores for DiffDock under varying inference conditions, showing the same values as Fig. 5.5. Removing the default temperature scaling (SDE-20 $T = 1$, light blue) yields calibration profiles nearly identical to those of the deterministic ODE-20 sampler (green), which sit slightly closer to perfect calibration. This confirms that DiffDock's default overconfidence (SDE-20, yellow) is inflated, partly by its applied temperature heuristics ($\tau > 1$) rather than by the dynamics of the stochastic differential equation itself.

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Declaration of authorship

I, Quentin Fuchs of Lucy Cavendish College, being a candidate for the Master of Philosophy in Data Intensive Science, hereby declare that this report and the work described in it are my own, unaided work, and that the report does not contain material that has already been used to any substantial extent for a comparable purpose. All direct or indirect sources used are acknowledged as references. I am aware that the thesis in digital form can be examined for the use of unauthorized aid and in order to determine whether the report as a whole or parts incorporated in it may be deemed as plagiarism. For the comparison of my work with existing sources I agree that it shall be entered in a database where it shall also remain after examination, to enable comparison with future theses submitted. I am willing to have my report made available to the students and staff of the University. Further rights of reproduction and usage, however, are not granted here.

In preparation for this report, I adhered to the coursewide AI policy. Claude Code assisted in understanding the existing codebase and in writing functions and shell scripts. Gemini and Claude were further used for drafting and proofreading of the report.

Cambridge, 06.07.2026

Quentin Fuchs